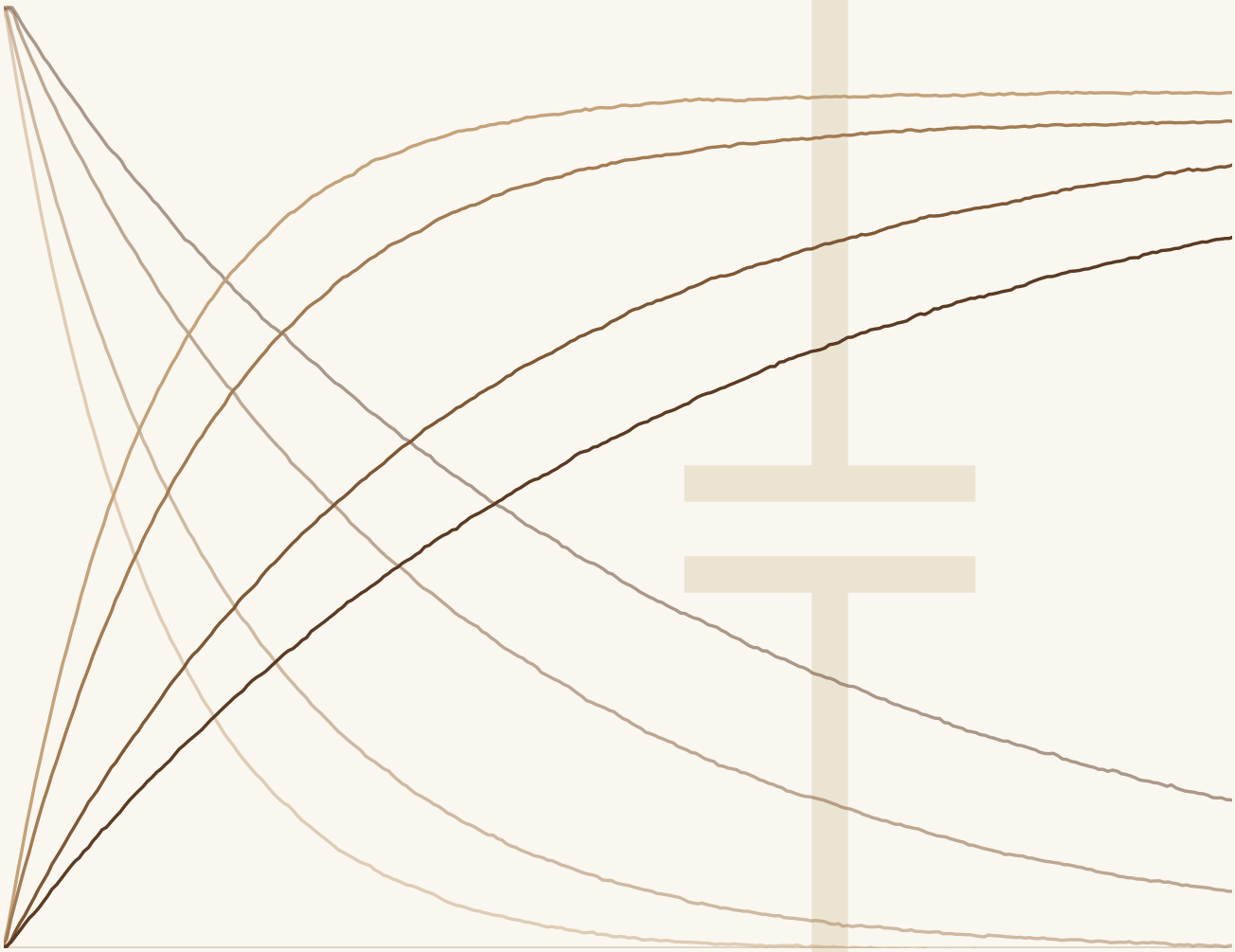
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RC Transient Response



ELT-1031 · Lab 06 · prof. Ashworth

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Part 1 Prelab Calculations

All four circuits charge through the same 1 MΩ from the same 15 V, so the time constant is set by the capacitance alone: 47 s, 100 s, 147 s and 32 s. Every circuit reaches 9.48 V after 1τ charging and falls to 5.52 V after 1τ discharging, and the current starts at 15 µA in all four.

1.1 Equivalent capacitance

Capacitors in parallel add, so Circuit 3 is 147 µF:

$$C_{eq} = 47\ \mu F + 100\ \mu F = 147\ \mu F \tag{1}$$

Capacitors in series combine the way parallel resistors do, so Circuit 4 is smaller than either capacitor:

$$C_{eq} = \frac{47\ \mu F \cdot 100\ \mu F}{47\ \mu F + 100\ \mu F} = 31.97\ \mu F \approx 32\ \mu F \tag{2}$$

1.2 Time constants and expected values

With $R = 1\ M\Omega$, each microfarad adds one second,[1] so τ in seconds equals the capacitance in µF:

$$\tau = R \cdot C = 1\ M\Omega \cdot 47\ \mu F = 47\ s \tag{3}$$

While charging, the capacitor voltage climbs toward V_S and gets 63.2 % of the way there in one time constant:

$$v_C(t) = V_S\left(1 - e^{-t/\tau}\right) \qquad v_C(\tau) = 0.632 \cdot 15\ V = 9.48\ V \tag{4}$$

While discharging, it falls from its starting voltage and has 36.8 % left after one time constant:

$$v_C(t) = V_0\, e^{-t/\tau} \qquad v_C(\tau) = 0.368 \cdot 15\ V = 5.52\ V \tag{5}$$

At $t = 0$ all 15 V sits across the 1 MΩ: when charging because the empty capacitor acts as a short, and when discharging because the full capacitor is the source. That is the maximum current, and it decays with the same τ either way:

$$I_{C,MAX} = \frac{V_S}{R} = \frac{15\ V}{1\ M\Omega} = 15\ \mu A \tag{6}$$

$$i_C(\tau) = 0.368 \cdot 15\ \mu A = 5.52\ \mu A \tag{7}$$

None of the voltages or currents depend on the capacitance; only the time it takes to reach them does (Table 1).

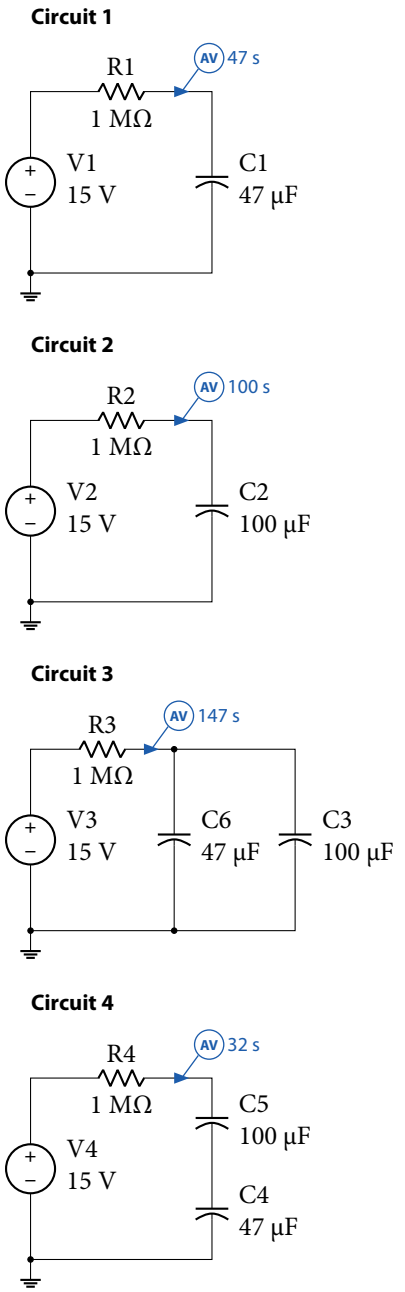


Figure 1. The four RC circuits from the Part 1 table, redrawn from the handout, with the AV probes from the Part 2 Multisim build. Each probe reads its node's voltage to ground and the current in the direction of its arrow, into the capacitor, and is named for its circuit's time constant.

[1] $1\ M\Omega \cdot 1\ \mu F = 10^{6}\ \Omega \cdot 10^{-6}\ F = 1\ s$.

	Circuit 1	Circuit 2	Circuit 3	Circuit 4
Configuration	1 MΩ + 47 µF	1 MΩ + 100 µF	1 MΩ + (47 µF ‖ 100 µF)	1 MΩ + (47 µF in series with 100 µF)
Capacitance	47 µF	100 µF	147 µF	31.97 µF
$\tau = R \cdot C$	47 s	100 s	147 s	31.97 s
Source	15 V	15 V	15 V	15 V
V_C after 1τ, charging	9.48 V	9.48 V	9.48 V	9.48 V
V_C after 1τ, discharging	5.52 V	5.52 V	5.52 V	5.52 V
Maximum I_C	15 µA	15 µA	15 µA	15 µA
I_C after 1τ	5.52 µA	5.52 µA	5.52 µA	5.52 µA
V_C after 5τ, charging	14.90 V	14.90 V	14.90 V	14.90 V

Table 1. Part 1 prelab values. The last two rows are not in the handout's table; Parts 2 and 3 use them.

Part 2 RC Transient Response (Charging)

In Multisim each capacitor reaches 63.2 % of 15 V at its own time constant. Circuits 1–3 match the calculated 9.48 V and 5.52 µA within 0.4 %; Circuit 4 reads 1.7 % low on voltage and 2.8 % high on current.

2.1 Multisim

Circuit 1 was built from the 15 V source, 1 MΩ and 47 µF, with an AV probe reading VDC and IDC and its arrow pointing into the capacitor. A Transient analysis ran to TSTOP = 1000 s with the initial conditions set to User-Defined.[2]

The current trace was moved to a right-hand axis (Figure 2, recreated in Figure 3). Circuits 2–4 were then built beside it with each probe renamed to its time constant (Figure 1), and the analysis was re-run plotting the four voltages (Figure 4) and then the four currents (Figure 5). Instead of reading values off the Grapher's cursors, each plot was exported to Excel, and the values at 1τ and 5τ were interpolated between the two nearest exported points.

[2] Multisim time period defaults were used, which turned out to be somewhat curious choices: the export doubles from 0.2 s up to 25.6 s, then holds 20 s apart.

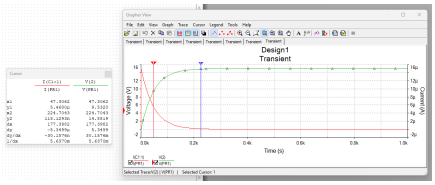


Figure 2. Multisim's Grapher with Circuit 1's two traces, before the export. Cursor 1 at 47.3 s reads 9.532 V and 5.468 µA.

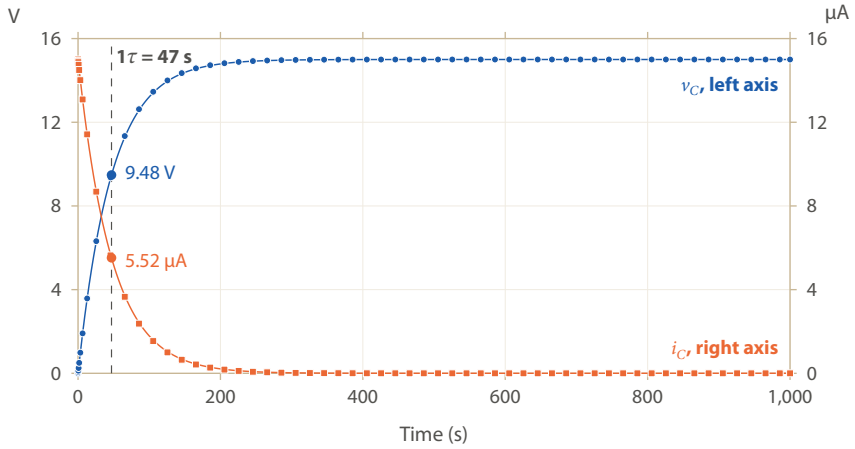


Figure 3. Circuit 1 charging: capacitor voltage and current, with 1τ = 47 s marked. Lines are calculated values; marks are the exported points.

2.2 Results

For Circuit 1 alone, the exported data at 47 s reads 9.502 V and 5.498 µA, within 0.4 % of the calculated 9.48 V and 5.52 µA. Circuits 2 and 3 agree as closely; Circuit 4 is the outlier (Table 2).

Circuit	τ (s)	V_C at 1τ (V)	Error (%)	I_C at 1τ (µA)	Error (%)	V_C at 5τ (V)
1	47	9.502	+0.23	5.498	−0.39	14.904
2	100	9.474	−0.07	5.526	+0.12	14.900
3	147	9.486	+0.06	5.514	−0.11	14.899
4	31.97	9.323	−1.66	5.677	+2.85	14.909

Table 2. Multisim values at 1τ and 5τ while charging, from the exported data. Errors are against the calculated 9.48 V and 5.52 µA; all four should read 14.90 V at 5τ.

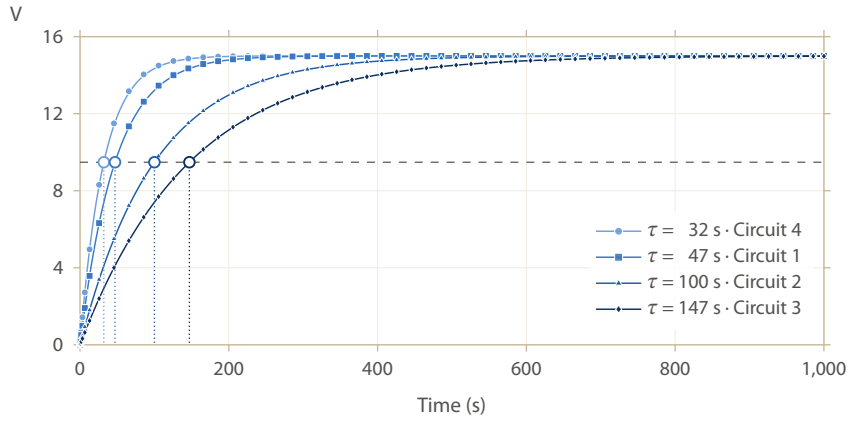


Figure 4. Capacitor voltage while charging, all four circuits. Each crosses 9.48 V at its own time constant.

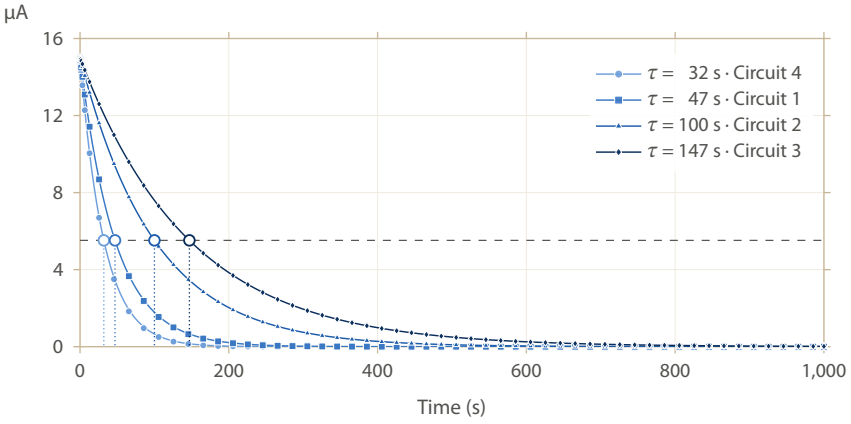


Figure 5. Capacitor current while charging. Every circuit starts at 15 µA and falls to 5.52 µA at its own time constant.

Circuit 4 charged fastest and Circuit 3 slowest, the same order as their time constants: 32, 47, 100, then 147 s (Figure 4). More capacitance takes longer to fill through the same 1 MΩ. All four reached the same voltage after 5τ, 14.90 V within 0.01 V; they just took between 160 s and 735 s to get there.

The currents decay in the same order (Figure 5). They're shown to be inversely proportional to the capacitor voltage, $(V_S - v_C)/R$. Once the capacitor is 63.2 % charged, the current is down to 36.8 % of its start, and every circuit starts at 15 µA because R is the same.

Circuit 4 has an outlying error but it's explained by the PicoScope export's point spacing. With TSTOP = 1000 s, Multisim saved points 20 s apart after the first

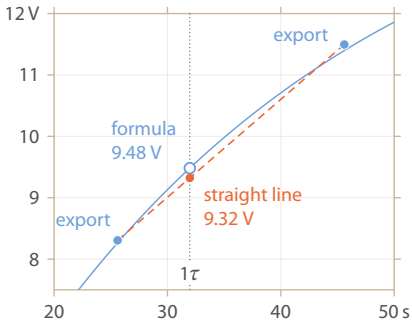


Figure 6. Circuit 4 near its 1τ. The export's nearest points are 20 s apart, and the straight line between them passes 0.16 V under the curve at 31.97 s.

25 s (25.6 s, 45.6 s, 65.6 s and so on), so the export has no point at Circuit 4's 1τ of 32 s. Its value there was interpolated between the points at 25.6 s and 45.6 s. The curve bends enough over those 20 s that the straight line falls 0.16 V under it (Figure 6). The saved points themselves are within 0.1 V of the formula. For Circuits 1–3 the same gap is a smaller fraction of τ, and the error stays under 0.4 %.

Part 3 RC Transient Response (Discharge)

With the sources removed and every capacitor starting at 15 V, each voltage falls to 36.8 % at its own time constant. The readings mirror Part 2: every 1τ voltage is 15 V minus its charging value, and Circuit 3 again takes the longest and Circuit 4 the shortest.

3.1 Multisim

The Part 2 file was saved as a copy, the four sources were removed and replaced with a short (Figure 7). An initial condition on each capacitor takes the place of the charging step: 15 V on Circuits 1–3, including both capacitors in Circuit 3, since parallel capacitors share the same voltage.

Circuit 4's two capacitors are in series, so each carries the full series charge Q_T, and the 15 V splits between them in inverse proportion to capacitance:

$$Q_T = C_{eq} \cdot V_S = 31.97\ \mu F \cdot 15\ V = 0.480\ mC \tag{8}$$

$$V_{100\ \mu F} = \frac{0.480\ mC}{100\ \mu F} = 4.80\ V \qquad V_{47\ \mu F} = \frac{0.480\ mC}{47\ \mu F} = 10.20\ V \tag{9}$$

The smaller series capacitor has a larger voltage drop, and the two add back to 15 V. The analysis was re-run to 1000 s, plotting the four voltages (Figure 8) and then the four currents (Figure 9). The probes were left as in Part 2, so the current now flows out of the capacitor against the arrow and reads negative.

3.2 Results

Circuit 3 took the longest to discharge and Circuit 4 the shortest (Figure 8), the same order as charging and for the same reason. More capacitance holds more charge at 15 V, and the same 1 MΩ drains it no faster than 15 µA.

The current curves rank the same way (Figure 9). With only the 1 MΩ resistor in the loop, the current is the capacitor voltage divided by 1 MΩ,[3] so each current curve is its voltage curve flipped negative. That is why the µA readings in Table 3 match the volt readings.

Each 1τ reading is the Part 2 value turned over, 5.498 V = 15 V − 9.502 V for Circuit 1 and likewise for the rest. Charging and discharging follow the same exponential, one rising toward 15 V and the other falling toward 0 V. Circuit 4's +2.85 % error comes from the same 20 s point spacing as in Part 2.

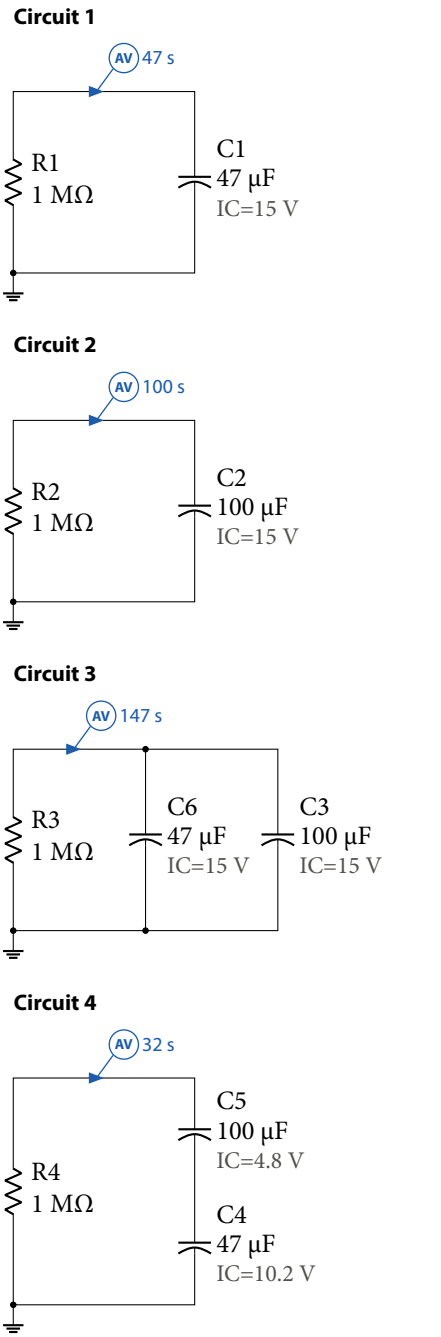


Figure 7. Part 3 circuits, redrawn from the Multisim build. The sources are gone, each 1 MΩ *sits across its capacitor, and the initial conditions (IC) stand in for the charge. Circuit 4's* 4.8 V *and* 10.2 V *add to* 15 V. *The AV probes are where they were in Part 2, so the discharge current flows against their arrows.*

[3] 1 V across 1 MΩ is 1 µA.

Its 5τ reading of 0.091 V is close to the expected 0.10 V. On a value that small, the percent error says more about the measurement technique.

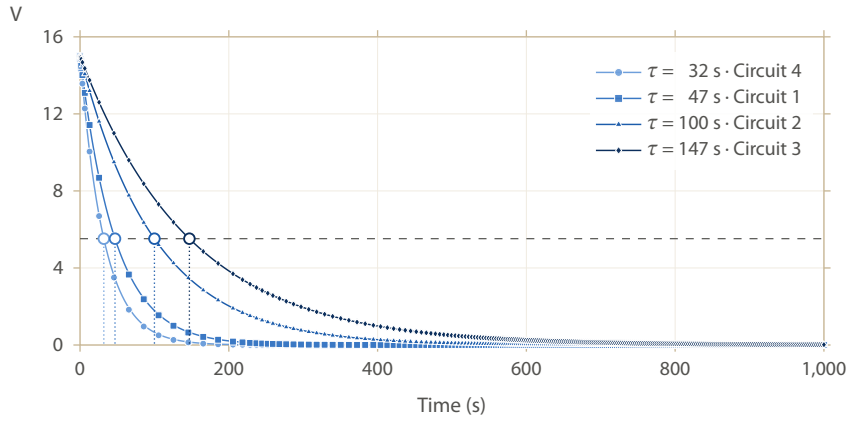


Figure 8. Capacitor voltage while discharging. Each falls to 5.52 V at its own time constant.

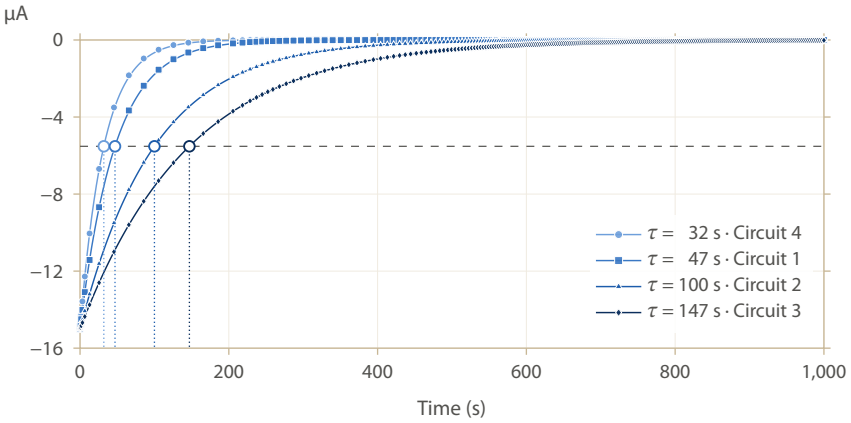


Figure 9. Capacitor current while discharging. Each starts at −15 μA, negative because the current now leaves the capacitor against the probe arrow.

Circuit	τ (s)	V_C at 1τ (V)	Error (%)	I_C at 1τ (μA)	Error (%)	V_C at 5τ (V)
1	47	5.498	−0.39	−5.498	−0.39	0.095
2	100	5.526	+0.12	−5.526	+0.12	0.100
3	147	5.514	−0.11	−5.514	−0.11	0.101
4	31.97	5.677	+2.85	−5.677	+2.85	0.091

Table 3. Multisim values at 1τ and 5τ while discharging, from the exported data. Errors are against the calculated 5.52 V and 5.52 μA, current by magnitude; all four should read about 0.10 V at 5τ.

Part 4 Hardware Verification of RC Transient Responses

On the breadboard each capacitor still crossed the magic 63.2 % figure close to its own $R \cdot C$. The scope captures put six of the eight time constants within 4.3 % under $R \cdot C$, and the discharges of Circuits 1 and 4 5.9 % and 3.3 % over it. Counting the meter's internal resistance, all of them should have come in 9.4 % under, so the hardware ran slower than that prediction. The stopwatch times ran 0.3–5.7 % long.

4.1 Calculation prep

On the DMM the supply read 15.008 V straight from the +15 V rail to ground, and 13.592 V through the 1 MΩ (Figure 10). The second reading is lower

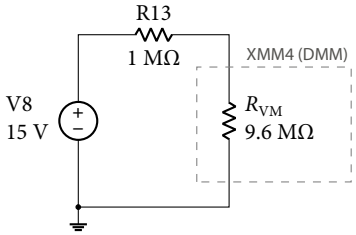


Figure 10. Step 5's measurement, redrawn from the handout with the DMM drawn as its internal resistance. R_{VM} and R13 form a voltage divider, so the meter reads only the voltage across R_{VM}, V_M, rather than that of the full 15 V supply.

because the DMM passes a small amount of current. Its internal resistance R_{VM} draws 1.42 µA through the 1 MΩ, which drops 1.42 V. The two resistors are a voltage divider, so VDR solved for R_{VM} gives the meter's resistance:[4]

$$V_M = V_S \frac{R_{VM}}{R_{VM} + R} \tag{10}$$

$$R_{VM} = R \frac{V_M}{V_S - V_M} = 1\ \text{M}\Omega \cdot \frac{13.592\ \text{V}}{15.008\ \text{V} - 13.592\ \text{V}} = 9.598\ \text{M}\Omega \tag{11}$$

The DMM stays across the capacitor for the rest of Part 4, so the same divider sets the highest voltage the capacitor can reach through the 1 MΩ:

$$V_{MAX} = V_S \frac{R_{VM}}{R_{VM} + 1\ \text{M}\Omega} = 15.008\ \text{V} \cdot \frac{9.598\ \text{M}\Omega}{10.598\ \text{M}\Omega} = 13.59\ \text{V} \tag{12}$$

This matches the step 5 reading, as the handout predicts. After one time constant the capacitor should reach 63.2 % of V_{MAX}, or 8.59 V, so the stopwatch was stopped at that voltage.

[4] R_{VM} calculated to 9.598 MΩ, which is close to the handout's hint of about 10 MΩ.

4.2 Charging

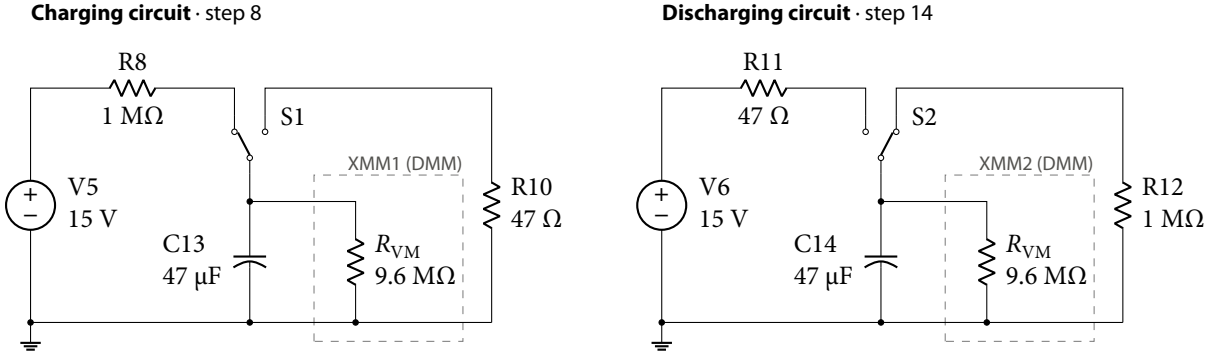


Figure 11. Part 4 circuits, redrawn from the handout (steps 8 and 14) with each DMM drawn as its internal resistance R_{VM} (step 15). Circuits 2–4 put their capacitors in C13's and C14's place.

The charging circuit from handout step 8 was wired with the 1 MΩ on one throw of the switch, the 47 Ω on the other, and the DMM across the capacitor (Figure 11, left). In one position the capacitor charges through the 1 MΩ; in the other the 47 Ω empties it for the next run, with a time constant of only 2.2 ms. For each circuit the switch was flipped as the stopwatch started, and the stopwatch was stopped when the DMM read 8.59 V. Each circuit was timed once (Table 4).

Every charge and discharge was also recorded on a PicoScope, on its ±50 V range at 50 s/div, or 100 s/div for Circuit 3, whose 5τ of 735 s would not fit on a 500 s screen. Instead of reading the scope's cursors, the CSV exports were analyzed: an exponential fitted to each capture gives the moment the switch flipped, the voltage the capacitor was heading for, and τ (Figure 12).[5]

For the captures the DMM was removed and the scope was connected across the capacitor through a ×10 probe, so the scope runs and the stopwatch runs

[5] With the capacitor shorted through the 47 Ω the scope read 0.41 V, so every trace is shifted down by that much. The ±50 V range moves in 0.46 V steps; averaging the samples in each time bin smooths them out.

were separate runs. The probe and the scope's input together are 10 MΩ, which loads the circuit almost exactly as the DMM's 9.6 MΩ does: V_{MAX} becomes 13.64 V instead of 13.59 V.

	Circuit 1	Circuit 2	Circuit 3	Circuit 4
Resistance	1 MΩ	1 MΩ	1 MΩ	1 MΩ
Capacitance	47 µF	100 µF	147 µF (parallel)	31.97 µF (series)
$\tau = R \cdot C$	47 s	100 s	147 s	31.97 s
Calculated V_{MAX}, charging	13.59 V	13.59 V	13.59 V	13.59 V
Calculated V_C after 1τ, charging	8.59 V	8.59 V	8.59 V	8.59 V
Measured τ, stopwatch to 63.2 %	49.66 s	104.5 s	154 s	32.07 s
Error, measured to calculated τ	+5.66 %	+4.50 %	+4.76 %	+0.30 %
Scope: time to 8.59 V	48.3 s	100.5 s	148.7 s	32.2 s
Scope: fitted τ	45.0 s	96.3 s	143.8 s	31.9 s
Scope: final voltage	13.1 V	13.2 V	13.3 V	13.5 V

Table 4. Part 4 charging data, the handout's table filled in. Errors are against $\tau = R \cdot C$, as the handout asks. The last three rows are from the scope captures and are not in the handout's table; the final voltage is where the fitted curve levels off.

4.3 Discharging

For the discharge runs the 47 Ω and 1 MΩ were swapped (Figure 11, right). Now the capacitor charges through the 47 Ω within milliseconds, and it discharges through the 1 MΩ slowly enough to time. In the discharge position the DMM's R_{VM} sits across the capacitor in parallel with the 1 MΩ, so the capacitor discharges through

$$R_{eq} = 1\ \text{M}\Omega \parallel R_{VM} = \frac{1\ \text{M}\Omega \cdot 9.598\ \text{M}\Omega}{1\ \text{M}\Omega + 9.598\ \text{M}\Omega} = 905.6\ \text{k}\Omega \tag{13}$$

and every time constant is 9.4 % shorter than $R \cdot C$:

$$\tau_{adj} = R_{eq} \cdot C = 905.6\ \text{k}\Omega \cdot 47\ \mu\text{F} = 42.57\ \text{s} \tag{14}$$

The discharges were not timed with the stopwatch; each measured τ in Table 5 is from its scope capture.

	Circuit 1	Circuit 2	Circuit 3	Circuit 4
Resistance	1 MΩ	1 MΩ	1 MΩ	1 MΩ
Capacitance	47 µF	100 µF	147 µF (parallel)	31.97 µF (series)
$\tau = R \cdot C$	47 s	100 s	147 s	31.97 s
Calculated V_C after 1τ, discharging	5.52 V	5.52 V	5.52 V	5.52 V
Adjusted $\tau = R_{eq} \cdot C$	42.57 s	90.56 s	133.13 s	28.96 s
Measured τ, scope to 36.8 %	49.8 s	96.8 s	143.8 s	33.0 s
Error, measured to adjusted τ	+16.95 %	+6.87 %	+8.03 %	+14.08 %
Error, measured to $R \cdot C$	+5.91 %	−3.21 %	−2.16 %	+3.34 %
Scope: starting voltage	14.9 V	14.9 V	14.9 V	14.9 V

Table 5. Part 4 discharging data, the handout's table filled in. Measured τ is from the scope captures. The last two rows are not in the handout's table.

In this circuit the capacitor charges to the full 15 V. The DMM forms a divider with the series resistor, and 47 Ω against 9.6 MΩ drops only 73 µV, where the 1 MΩ dropped 1.42 V. The scope agrees: before every discharge the capacitor sat at 14.9 V, within one 0.46 V step of the 15.008 V supply.

The meter's resistance still matters for the discharge, just not for its starting voltage. Once the switch flips, R_{VM} is a second path for the charge to drain through, in parallel with the 1 MΩ, and it shortens τ by 9.4 %. The 36.8 % target does not change, because it is a fraction of the starting voltage, and that is the full 15 V.

4.4 Results

Figure 12 puts all eight captures on one page, each row's time axis scaled to its own $R \cdot C$ so the 1τ lines line up. Circuit 4 again charged fastest and Circuit 3 slowest, the same order as Parts 2 and 3, and on the scaled axes all eight curves have the same shape.

The meter belongs in the charging prediction as much as the discharging one. While charging, the capacitor sees the 1 MΩ and R_{VM} as a Thevenin resistance of the same 905.6 kΩ, charging toward V_{MAX} instead of 15 V. So both tables should be held to the adjusted τ, and every capture came out slower than it, by 5.6–10.2 % (14.1 % for Circuit 4's discharge and 17 % for Circuit 1's).[6] Since R_{VM} was measured, the rest is most likely the parts themselves: electrolytic capacitors are usually rated ±20 %, and capacitance 6 to 10 % over the marked value would cover all but the discharges of Circuits 1 and 4. The 1 MΩ's ±5 % could account for some of it too. Neither part was measured on its own, so the two can't be separated.

The stopwatch errors are larger and all positive because the stopwatch was stopped at 8.59 V, 63.2 % of the calculated V_{MAX}. On the scope the charging curves level off lower, at about 13.1 to 13.5 V, below even the 13.64 V the probe alone allows. So 8.59 V is 64 to 66 % of the way and arrives 1 to 7 % after τ. Electrolytic capacitors leak a little current, which acts like one more resistor across the capacitor and pulls the divider's output down. Although they were separate runs, the stopwatch and the scope agree on when the capacitor reached 8.59 V within about 4 %. Near that point Circuit 3's curve climbs only 0.03 V per second, so a tenth of a volt between the two runs or the two instruments moves the time by three seconds.

Circuit 1 is the odd one: it discharged 10.7 % slower than it charged, where Circuits 2 and 3 agree within 0.5 % and Circuit 4 sits in between at 3.5 % (Figure 13). It can only be further understood with more trials.

[6] During the captures the probe's 10 MΩ stood in for R_{VM}, making the resistance 909 kΩ instead of 905.6 kΩ. Against that the captures run 5.2–9.8 % slow, and 13.7 % and 16.5 % for the discharges of Circuits 4 and 1.

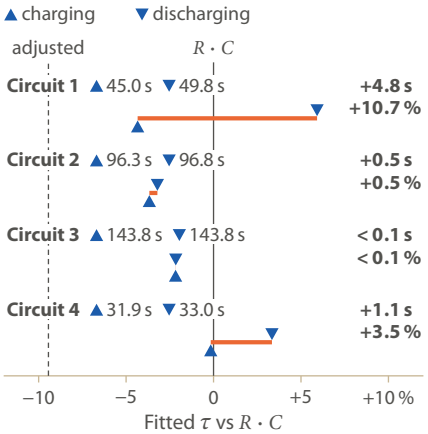


Figure 13. Each circuit's fitted τ from the scope, charging against discharging, measured from $R \cdot C$ (solid) with the meter-adjusted τ dashed. The orange bar is the gap between the two; the tags give how much longer the discharge took, in seconds and percent.

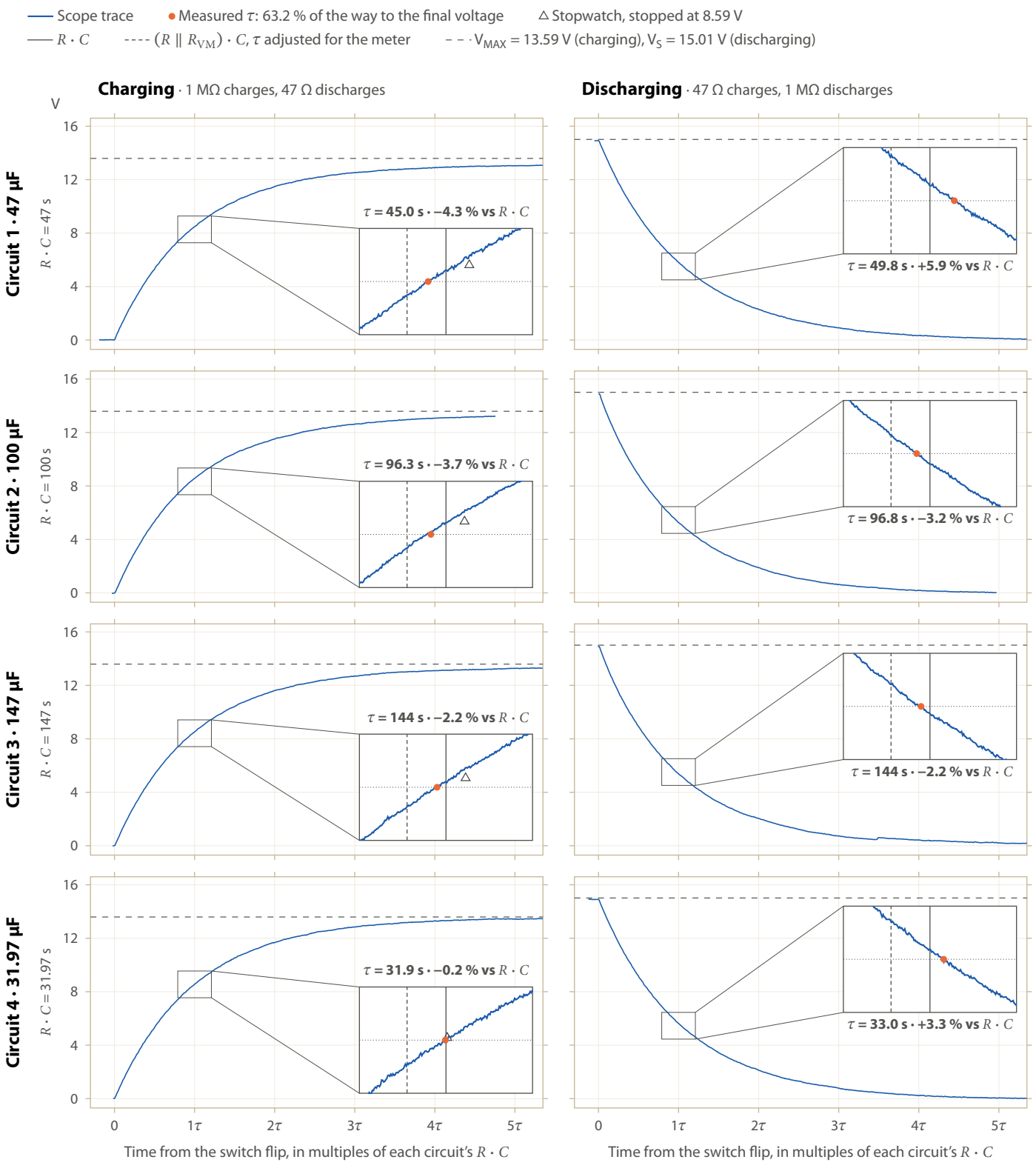


Figure 12. Every Part 4 scope capture, shifted down by the scope's 0.41 V *zero offset and plotted against time from the switch flip in multiples of each circuit's* $R \cdot C$*, so the* 1τ *grid lines line up down the page. Six of the eight captures reach their* 63.2 % *point between the meter-adjusted* τ *and* $R \cdot C$*, nearer* $R \cdot C$*; only the discharges of Circuits 1 and 4 run past it. Insets magnify the* 1τ *crossing five times in time. The scope was not triggered by the switch, so each flip time is taken from the capture's fitted curve.*

Conclusion

Q1 State which charges faster, a small capacitor or a large capacitor. Describe the effect of putting capacitors in parallel or series on the charging rate and total capacitance in the circuit.

The small capacitor charges faster. Through the same resistor $\tau = R \cdot C$ grows with C, so the 32 µF series pair reached 63.2 % in 31.9 s on the scope and the 147 µF parallel pair took 143.8 s. Capacitors in parallel add, so the combination is larger than either one and charges more slowly. Capacitors in series combine the way parallel resistors do, so the combination is smaller than either one and charges faster. The time constants measured on the bench combine the same way: 45.0 s + 96.3 s = 141.3 s against 143.8 s measured in parallel, and the series formula gives 30.7 s against 31.9 s measured.

Q2 What effect does the time constant have on either charging or discharging? What effect would changing the value of resistance have on the time constant?

The time constant scales the capacitor's rate of approach to 100 % charge. After 1τ it is 63.2 % charged or has 36.8 % left, whatever τ is, and after 5τ it is within 0.7 % of the end value. A larger τ stretches the same curve out in time, which is why all eight captures look alike in Figure 12 once time is divided by $R \cdot C$. Since $\tau = R \cdot C$, τ is proportional to R: doubling the resistance doubles it.

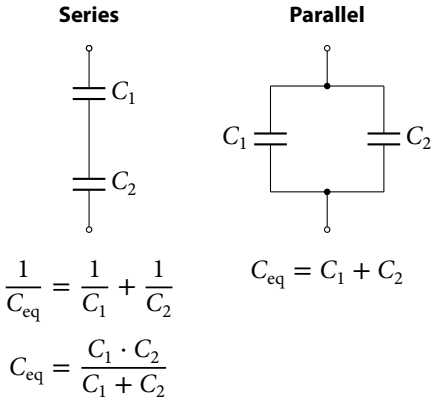


Figure 14. C_{eq} formulas for capacitors in both series and parallel. Both formulas (except for the series capacitance shortcut shown) are associative and commutative, and can therefore be repeated for an arbitrary number of elements.

Errors and Difficulties

Each circuit was timed once with the stopwatch, so there is no spread to show how repeatable the times are, and the scope captures were also one run each. Multiple trials would generally shed more certainty on the experiment.

The 1 MΩ charge/discharge resistor suits the stopwatch procedure, which needs time constants long enough to time by hand, but the scope procedure actually used could have employed a much smaller value. With 10 kΩ, every τ would be under 1.5 s, a full charge and discharge of Circuit 3 would take about 15 s instead of 25 min, and many more trials would fit in the same lab time.

The scope's ±50 V range moves in 0.46 V steps, about 3 % of the signal, and it read 0.41 V with the capacitor shorted. Averaging and the offset correction take care of both for τ, which depends only on the curve's shape, but absolute voltages read from the scope, like the final voltages in Table 4, are only good to a couple of tenths of a volt. A smaller range would have given finer steps. The scope was also not triggered by the switch, so the flip time had to be found from each trace.

Appendix Multisim Screenshots

The Multisim originals of Figures 1 and 7. Multisim default component names were unchanged; the redrawn figures copy all exactly.

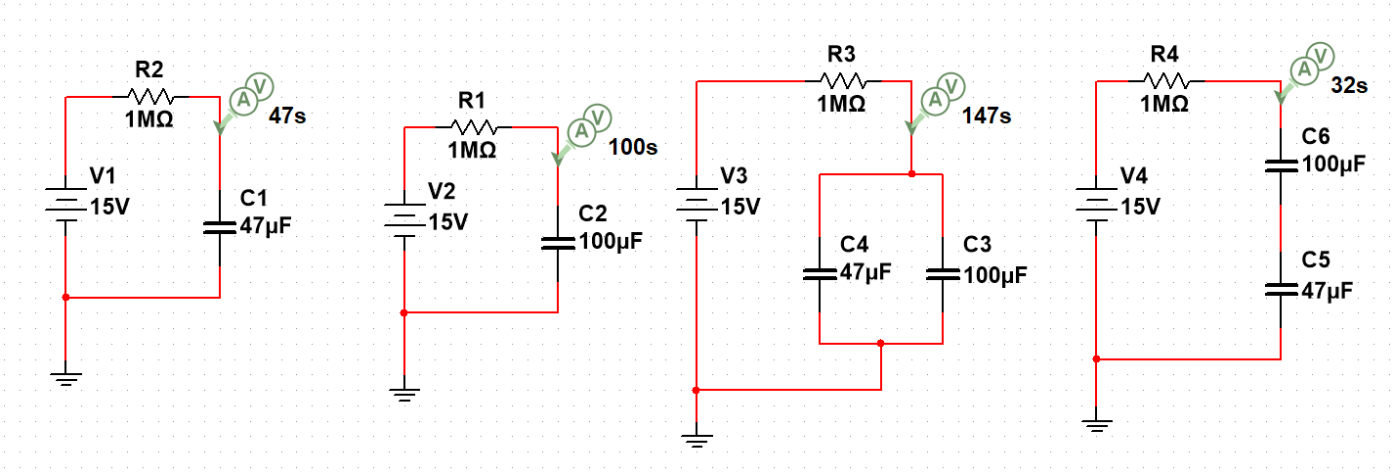


Figure 15. Part 2 circuits in Multisim. Each AV probe is named for its circuit's time constant and points into the capacitor.

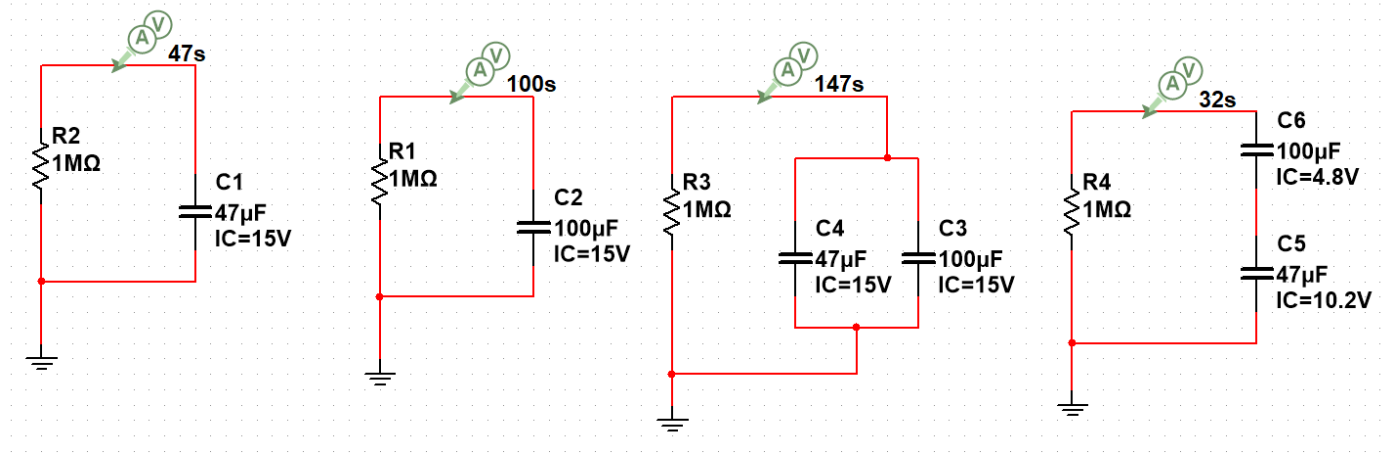


Figure 16. Part 3 circuits in Multisim, with the initial conditions (IC) on each capacitor.